Bayesian Data Analysis of the German Tank Problem

Theoretical Properties, Simulation Benchmarks, and Real-World Stress-Testing

Inferring Discrete Ceilings, Decision Theory, and Operational Assumption Violations

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Historical Context: Statistical Intelligence in WWII

Objective: Infer the unknown total number of tanks produced, N, using a small sample of observed serial numbers.

1. Espionage Inflation

Conventional military intelligence relied on spy networks and captured prisoners, drastically overestimating production by **4.5–8.2**$\times$ (1,000–1,550 tanks/mo).

2. Order Statistics

Allied statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–50), mathematically estimating production at **169–327** tanks/mo.

3. Speer Ministry Ledgers

Post-war Albert Speer production records confirmed statistical estimates converged within **4%–10%** of actual production.

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	**169**	122	+38.5%
June 1941 (Panzer III/IV)	1,550	**244**	271	−**9.9**%
August 1942 (Panzer III/IV)	1,550	**327**	342	−**4.4**%

Core Inferential Objective: *Infer the unknown population ceiling N, moving beyond point estimates to complete uncertainty quantification.*

Mathematical Formulation: Data, Parameter, and Likelihood

Explicitly separating what is observed from the unknown parameter we seek to infer.

The Data-Generating Mechanism

Serials drawn by simple random sampling without replacement:

$$S_1, \ldots, S_k \overset{\text{SRSWOR}}{\sim} \{1, 2, \ldots, N\}$$

- **Observed Data:** $D = \{s_1, \ldots, s_k\}$ with sample size k.
- **Order Statistics:** $1 \leq s_{(1)} < s_{(2)} < \cdots < s_{(k)} = m$.
- **Sample Maximum:** $m = \max(s_1, \ldots, s_k)$.
- **Physical Boundary:** $N \geq m$ is an absolute physical constraint.

Inferential Target & Unordered Likelihood

- **Inferential Target:** The target is the unknown population ceiling $N \in \mathbb{N}_{\geq m}$, *not the individual serial numbers*.
- The serial numbers D are observed data; N is the unknown quantity we want to learn about.

$$P(D \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m)$$

- Likelihood is strictly zero for all $N < m$.
- Sample maximum m captures all sample dependence on N.

Key Insight: *The serial numbers are observed evidence; the population ceiling N is the unknown inferential target.*

Classical Solution: The UMVUE & Its Limitations

Frequentist point estimation achieves repeated-sampling unbiasedness but provides no posterior distribution.

The UMVUE Estimator

By the likelihood ratio criterion for minimal sufficiency (Bahadur, 1954), $M = \max_i S_i$ is complete and minimal sufficient.

$$\widehat{N}_{\mathrm{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1$$

$$\mathrm{Var}(\widehat{N}_{\mathrm{UMVUE}}) = \frac{(N-k)(N+1)}{k(k+2)}$$

- Unbiased across hypothetical repeated samples ($\mathbb{E}[\widehat{N}] = N$).
- Simple closed-form computation requiring only m and k.

Fundamental Limitations of Classical Approach

- **No Posterior Uncertainty Distribution:** The classical approach constructs a point estimator that is unbiased over repeated samples. It does not provide a posterior distribution describing uncertainty about N given this specific sample.
- **Boundary Violations:** Asymptotic Wald intervals ($\widehat{N} \pm 1.96 \cdot \mathrm{SE}$) extend below m, placing mass on physically impossible values ($N < m$).
- **Inflexible to Decision Costs:** Cannot adapt to asymmetric operational losses where underestimation carries disastrous strategic costs.

Transition to Bayes: *Point estimators provide only an average; Bayesian inference constructs the full posterior distribution $P(N \mid D)$.*

SECTION 01

Theoretical Properties & Bayesian Foundations

Inferring the posterior distribution, reference priors, loss geometry, and boundary admissibility

Bayesian Inference: What Are We Trying to Infer?

Establishing the inferential objective: moving from raw identifiers to an epistemic probability distribution.

1. What Do We Know?

- The observed sample of serial numbers $D = \{s_1, \ldots, s_k\}$.
- The sample maximum $m = \max(s_1, \ldots, s_k)$.
- Physical fact: the population must satisfy $N \geq m$.

2. What Is Unknown?

- The true population ceiling $N \in \{m, m+1, m+2, \ldots\}$.
- We do not know how far above m the true ceiling lies.

3. What Does Bayesian Inference Give Us?

A posterior distribution describing the probability of every possible population size, conditional on the observed data:

$$\boxed{P(N = n \mid D)}$$

4. What Do We Do With This Distribution?

- Quantify uncertainty via exact credible intervals.
- Derive optimal point estimates tailored to specific loss functions.
- Make principled decisions accounting for risk asymmetry.

Inferential Pipeline:

Observed Data $D = \{s_1, \ldots, s_k\}$ $\longrightarrow$ **Posterior Distribution** $P(N = n \mid D)$ $\longrightarrow$ Estimation & Decision Rules $\widehat{N}_L$

Core Message: *The posterior distribution itself is the primary inferential output; point estimates are derived afterwards.*

Ingredients of Bayesian Inference: Likelihood, Prior, Posterior

How observed evidence combines with prior assumptions to quantify uncertainty about N.

1. Likelihood: Data Evidence

What does the sample tell us about N?

$$P(D \mid N = n) = \frac{1}{\binom{n}{k}} \mathbb{I}(n \geq m)$$

- Measures the probability of observing D under each candidate n.
- Equals **zero for** $n < m$: automatically assigns zero posterior probability to physically impossible population sizes.

2. Prior: Initial Assumptions

What do we assume about N before observing data?

- **Flat Prior ($\pi \propto 1$):** Assumes equal prior weight across all candidate integers.
- **Reciprocal Prior ($\pi \propto 1/n$):** Discrete analog of scale invariance; assigns equal mass to multiplicative decades/octaves ($\sum_{A}^{cA} \frac{1}{n} \approx \log c$).

3. Posterior: Updated Belief

What do we believe about N after observing data?

$$P(N = n \mid D) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- Combines serial evidence with prior structure.
- Quantifies epistemic uncertainty across all physically admissible $n \geq m$.

Posterior Interpretation: *The posterior combines sample data with prior assumptions to assign exact probabilities to all candidate population sizes.*

Discrete Posteriors: Propriety & Moment Existence Thresholds

Different prior assumptions produce different posterior distributions, even when the observed data remain unchanged.

Exact Discrete Posterior PMFs ($n \geq m$)

- **Flat Prior ($\pi \propto 1$):**

$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}$$

Normalizing constant telescopes exactly via negative hypergeometric summation.

- **Reciprocal Prior ($\pi \propto 1/n$):**

$$P(N = n \mid m, k) = \frac{1}{Z(m,k)} \frac{1}{n\binom{n}{k}}$$

$Z(m,k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}$ converges for all $k \geq 1$ (no simple closed form; evaluated numerically).

Moment Existence Thresholds & Pareto Limit

Prior Distribution	Proper Posterior	Finite Mean $\mathbb{E}[N]$	Finite Var $\mathrm{Var}(N)$
Flat Prior ($\pi \propto 1$)	$k \geq 2$	$k \geq 3$ ($\mathbb{E} = 2m - 2$)	$k \geq 4$
Reciprocal Prior ($\pi \propto 1/n$)	$k \geq 1$	$k \geq 2$ ($\mathbb{E} \approx 2m$)	$k \geq 3$

Continuous Pareto Limit ($N \gg k$):

$$p(N \mid m, k) \approx \frac{(k-1)m^{k-1}}{N^k}, \quad N \geq m$$

- Cumulative CDF: $F(N) = 1 - (m/N)^{k-1}$.
- Median: $\widehat{N}_{\mathrm{med}} \approx m \cdot 2^{1/(k-1)}$.

Theoretical Takeaway: *Reciprocal prior guarantees posterior propriety for all $k \geq 1$, whereas flat prior requires $k \geq 2$ for propriety and $k \geq 3$ for finite mean.*

Estimating N from a Loss Function: Bayesian Decision Theory

A posterior distribution describes uncertainty, but a decision requires us to select a single estimate.

The Decision-Theoretic Framework

- The optimal estimate depends on the cost assigned to different estimation errors.
- Let $a \in \mathbb{N}_{\geq m}$ denote an estimation action.
- Loss function $L(N, a)$ quantifies the consequence of predicting a when true population is N.
- **Posterior Expected Loss (Risk):**

$$R(a) = \mathbb{E}[L(N, a) \mid D] = \sum_{n=m}^{\infty} L(n, a) P(N = n \mid D)$$

- **Optimal Bayes Estimator:**

$$\widehat{N}_L = \arg\min_a R(a)$$

Example: Squared-Error Loss (L_2)

When estimation errors are penalized quadratically:

$$L(N, a) = (N - a)^2$$

$$R(a) = \mathbb{E}[(N - a)^2 \mid D] = \mathrm{Var}(N \mid D) + (\mathbb{E}[N \mid D] - a)^2$$

Setting the derivative to zero yields:

$$\boxed{\widehat{N}_{L_2} = \mathbb{E}[N \mid D] \quad \text{(Posterior Mean)}}$$

- The posterior mean is not "the" Bayesian estimate; it is optimal specifically for squared-error loss.
- If errors carry asymmetric costs, a different estimator is required.

Core Rule: *Decisions follow inference: point estimators cannot be chosen without defining a loss function.*

Loss Functions & Bayesian Estimators: Matching Risk to Mission

Point estimators shift systematically with the decision-maker's loss geometry.

1. 0–1 Loss

$L(N, a) = \mathbb{I}(a \neq N)$

$\widehat{N} = \text{Mode} = m$

- Selects most probable single value.
- Severe negative bias (-16.7% at $k = 5$).

2. Absolute (L_1)

$L(N, a) = |N - a|$

$\widehat{N} = \text{Median}$

- Linear penalty on deviations.
- Robust against heavy Pareto tails (-0.9% bias).

3. Squared (L_2)

$L(N, a) = (N - a)^2$

$\widehat{N} = \mathbb{E}[N \mid D]$

- Quadratic penalty on large errors.
- Pulled rightward by skewed Pareto tail.

4. Check Loss L_q

$L_q = (N - a)(q - \mathbb{I}(N < a))$

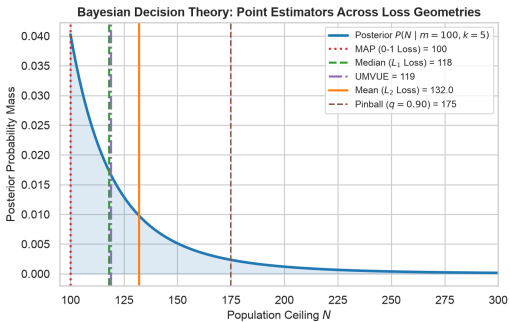
$\widehat{N} = F^{-1}(q \mid D)$

- $q = 0.5$: symmetric median.
- $q = 0.9$: undercounting penalized $9\times$ worse $\Longrightarrow$ defensive safety buffer.

Military Application: *In defense intelligence, underestimating enemy armor risks catastrophic defeat; asymmetric check loss ($q = 0.90$) provides a principled operational buffer.*

Numerical Illustration: Sensitivity of Estimates to Loss

Illustration using the flat-prior posterior, with sample maximum $m = 100$ and sample size $k = 5$.



Point Estimates Across Objectives ($m = 100, k = 5$)

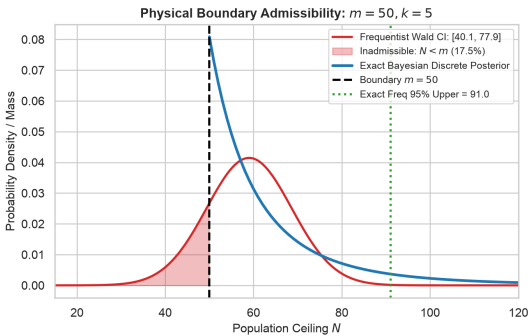
Objective / Loss	Estimator	Estimate	Strategic Meaning
0–1 Loss	MAP / Mode	**100**	Extreme lower bound; severe undercount
L_1 Absolute Loss	Posterior Median	**118**	Minimizes expected absolute error
L_2 Squared Loss	Posterior Mean	**132**	Quadratic balance; tail-inflated
Check Loss ($q = 0.90$)	90th Percentile	**175**	+75% defensive safety buffer

- Under the same observed data ($m = 100, k = 5$), optimal estimates span from 100 to 175.
- Posterior expected loss is minimized at a different point for each loss function.

Key Finding: *The same observed data can produce vastly different optimal estimates depending on how we value overestimation and underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian Intervals

The physical constraint $N \geq m$ must be respected by any valid uncertainty interval.



Frequentist Wald Boundary Collapse

Wald interval $\widehat{N} \pm 1.96 \cdot \mathrm{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$. Asymptotic Gaussianity fails completely near the support boundary.

Approximate Frequentist Pivotal CI

Large-N pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the physical boundary by construction.

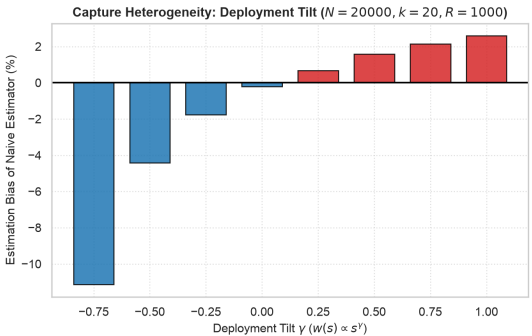
Bayesian Credible Coherence

Reciprocal-prior 95% credible interval strictly enforces physical admissibility ($P(N < m \mid D) = 0$) and approximately matches the pivotal bound $[50.0, 91.0]$.

Key Finding: *The reciprocal-prior posterior respects the physical lower bound and produces credible intervals that approximately agree with the large-N frequentist pivotal interval.*

Sensitivity to Non-Uniform Sampling: When the Maximum Misleads

Evaluating estimator robustness when capture probability varies with serial age: $w(s) \propto s^{\gamma}$.



Weighted Capture Mechanics

- $\gamma > 0$ (newer tanks prioritized): pushes $m \to N$, inducing $+1.6\%$ bias at $\gamma = +0.5$.
- $\gamma < 0$ (survivorship / depot attrition): induces -4.4% bias at $\gamma = -0.5$ and -11.1% bias at $\gamma = -0.75$.

Sufficiency Structure Breakdown

- **Known γ:** m remains sufficient for N via tilted factor $g_N(m; \gamma)$.
- **Unknown γ:** m alone ceases to be sufficient; joint minimal sufficient statistic is $(m, \sum_{i=1}^{k} \log s_i)$.

Methodological Warning: *Applying standard estimators when sampling is tilted introduces systematic bias; joint inference over (N, γ) is required.*

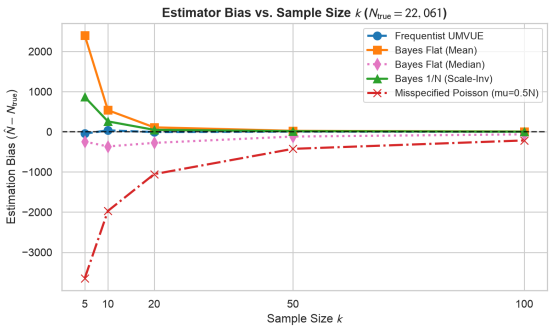
SECTION 02

Controlled Monte Carlo Simulation Study

Evaluating bias, RMSE contraction, interval calibration, and prior sensitivity across $R = 4,000$ trials ($N = 20,000$)

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$.



Frequentist UMVUE Unbiasedness

$\widehat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all k (-48.0 ± 54.6 at $k = 5$, $+5.2 \pm 6.0$ at $k = 50$). Unbiased over hypothetical repeated samples.

Bayes Reciprocal Prior

Posterior Mean has modest bias ($+767.0 \pm 56.8$ at $k = 5$), rapidly decaying to $+13.2$ at $k = 50$. Posterior Median exhibits negative shrinkage (-901.5 at $k = 5$).

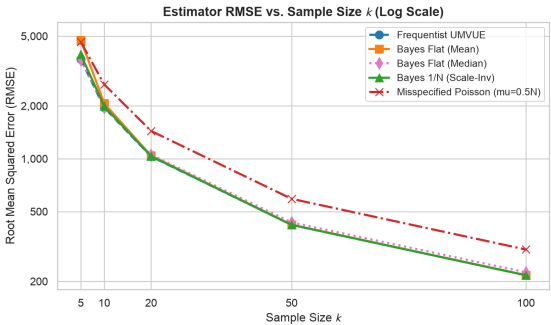
Flat Prior Tail Inflation

Skewed Pareto tail inflates flat posterior mean by $+2,168.6 \pm 60.7$ at $k = 5$.

Core Distinction: *Unbiasedness is a repeated-sampling frequentist property; Bayesian posterior summaries are chosen using the posterior distribution and a decision objective.*

Faster-Than-Standard-CLT Error Contraction: RMSE & Variance

Boundary order-statistic structure accelerates Root Mean Squared Error decay to $\mathcal{O}(k^{-1})$.



Theoretical Error Rate

Because N bounds the support, the UMVUE variance satisfies:

$$\mathrm{Var}(\widehat{N}) = \frac{(N-k)(N+1)}{k(k+2)} = \mathcal{O}(k^{-2})$$

giving accelerated rate RMSE $= \mathcal{O}(k^{-1})$, far outperforming standard CLT rate $\mathcal{O}(k^{-1/2})$.

Empirical Concordance

A $10\times$ increase in k ($5 \to 50$) yields a $\mathbf{9.0}\times$ error reduction ($3,455.2 \to 382.5$), closely matching theoretical $8.6\times$ factor and asymptotic $10\times$ scaling.

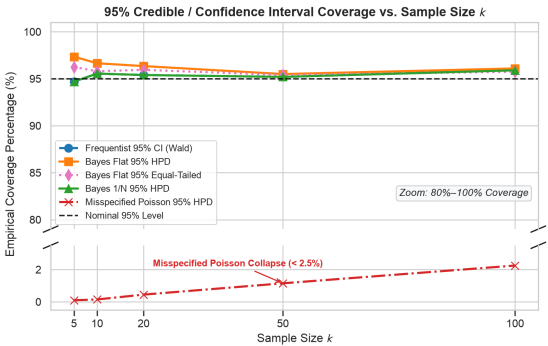
Estimator Convergence

For $k \geq 50$, all point estimators achieve identical precision (≈ 383).

Contraction Rule: *The order-statistic boundary structure drives an accelerated $\mathcal{O}(k^{-1})$ RMSE decay, reducing error by nearly an order of magnitude as sample size increases tenfold.*

Prior Sensitivity: Calibration and the Informative Prior Trap

Evaluating interval reliability and the catastrophic consequences of dogmatic prior misspecification.



Reciprocal Prior Calibration

Empirical coverage closely tracks nominal 95%: **94.4%** $\pm$ **0.36%** at $k = 5$ and **95.6%** $\pm$ **0.32%** at $k = 50$.

Informative Boundary Trapping

A dogmatically narrow misspecified Poisson($\mu = 10k$) prior traps posterior against the boundary $[m, m + 5]$, collapsing coverage to **0.3%**.

Hierarchical Hyperprior Resolution

Gamma hyperprior over Poisson rate ($\mathrm{CV} \geq 1$) provides adaptive dispersion, restoring coverage to **94.6%–95.0%** even under 50% prior mean error.

Calibration Finding: *The reciprocal prior produces empirical coverage close to 95% in our simulation. A misspecified Poisson prior causes severe collapse, while a hierarchical prior restores calibration.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Multi-dimensional evaluation across $R = 4,000$ replications on $N_{\text{true}} = 20,000$ (Seed 42).

Framework	Estimator	k	Mean Bias $\pm$ SE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Approximate Pivotal CI	Large-N Pivotal Bound	5	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 60.7$	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.1	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal $(1/n)$	Posterior Mean	5	$+767.0 \pm 56.8$	3,671.1 / 3,618.6	**94.4%** $\pm$ 0.36%	13,622
Bayes Reciprocal $(1/n)$	Posterior Median	5	-901.5 ± 52.3	**3,427.5** / —	**94.4%** $\pm$ 0.36%	13,622
UMVUE (Frequentist)	$\widehat{N} = m(1 + 1/k) - 1$	50	$+5.2 \pm 6.0$	382.5 / 391.8	95.1% $\pm$ 0.34% (Wald)	1,536
Approximate Pivotal CI	Large-N Pivotal Bound	50	—	—	**95.6%** $\pm$ 0.32%	1,211
Bayes Reciprocal $(1/n)$	Posterior Mean	50	$+13.2 \pm 6.0$	382.8 / 392.0	**95.6%** $\pm$ 0.32%	1,209
Bayes Flat Prior	Posterior Mean	50	$+21.6 \pm 6.1$	383.4 / 392.4	**95.9%** $\pm$ 0.31%	1,237

1. Point Accuracy

UMVUE is exactly mean-unbiased; Bayes reciprocal median achieves lowest small-sample RMSE $(3,427.5)$.

2. Interval Reliability

Reciprocal credible interval achieves 94.4% coverage; Wald interval suffers from boundary distortion (90.0%).

3. Interval Precision

Reciprocal intervals are substantially sharper $(13,622)$ than flat prior intervals $(18,535)$.

Synthesis: *The reciprocal prior provides a useful balance between estimation accuracy, empirical calibration, and physical admissibility in the simulated setting.*

SECTION 03

Real-World Stress-Testing

Does the German Tank Problem still work when we apply it to real-world data?

The Reality Check: Two Real-World Datasets

Why we move from textbook simulations to physical operational data.

1. Archival WWII German Tank Records

- **Data Source:** Waffenamt ordnance records and Speer Ministry production registers.
- **Physical Artifacts:** Sequential serial numbers stamped on physical tank chassis and gearbox castings.
- **Historical Benchmark:** Validates that discrete order statistics successfully estimated production ceiling within -4.4%.

2. Modern Enterprise Invoices (E-Commerce)

- **Data Source:** 541,909 transaction line items across 4,372 customer accounts.
- **Sequential Identifiers: 22,061 distinct sales invoice numbers**.
- **Crucial Distinction:** 541,909 represents multi-item line purchases; 22,061 represents the distinct sequential invoice numbers.

The Core Question: We are not claiming commercial retail behaves like wartime manufacturing. We use these datasets as an **empirical stress-test** to ask: *what happens to our estimators when textbook assumptions fail?*

Investigation Goal: *Examine model assumptions against reality: how numbers are generated, missing values, and what we are counting.*

Assumption 1: How Identifiers Are Created (Fixed Hat vs. Flow of Time)

A random sample from a fixed population is fundamentally different from a stream generated over time.

The Textbook Model: "The Fixed Hat"

- Assumes a **static, complete population** $\{1, 2, \ldots, N\}$.
- All N objects exist simultaneously before observation begins.
- An observer draws a simple random sample of k numbers.
- Every serial number has an equal probability of capture.

Operational Reality: "The River Stream"

- Numbers are created sequentially as transactions occur.
- **Changing Invoice-Generation Intensity:**
  - Q1 Baseline: $\approx$ **370** invoices/week.
  - Q4 Acceleration: $>$ **790** invoices/week.
  - Peak Holiday Surge: **894** invoices in mid-November.
- **Why This Matters:** Identifiers reflect seasonal activity; later numbers appear only late in the year.

Statistical Lesson: A static population model cannot automatically be applied to a live stream where generation intensity changes over time.

Takeaway: *Textbook sampling draws from a completed jar; real operations observe an ongoing process that speeds up and slows down.*

Assumption 2: Are Numbers Contiguous? (The Missing Number Mystery)

The textbook assumes continuous numbering $1, 2, \ldots, N$. Real data has massive integer voids.

The "Numbered Slots" Analogy

Imagine a row of 45,223 numbered parking slots:

Slot Address Span: $[536, 365 \text{ to } 581, 587] \implies \mathbf{45,223}$ slots

- **Occupied Slots (Active Sales):** $\mathbf{22,061}$ (48.8%).
- **Empty Slots (Missing Numbers):** $\mathbf{23,162}$ (**51.2%** void rate!).
- *More than half of the numerical address slots have no completed sales invoice!*

Investigating the 23,162 Missing Numbers

- **1. Verified Cancellations (3,836 numbers):** Exactly 3,836 records bear a 'C-' prefix in raw logs. These represent verified customer cancellations (**16.6% of voids**).
- **2. Operational Hypotheses (19,326 numbers):** In the absence of server logs, plausible hypotheses include: abandoned shopping carts, payment gateway timeouts, and database sequence caching blocks.
- **Critical Distinction:** 3,836 are verified; the remaining 19,326 remain plausible operational hypotheses, not verified findings.

Historical Parallel: In WWII, Waffenamt pre-allocated chassis blocks to plants; when chassis were diverted to assault guns (StuG III/IV), voids appeared in standard tank sequences.

Takeaway: *Missing serial numbers are not random corruption; they reflect cancellations, system architecture, and block allocations.*

Assumption 3: Are Numbers Evenly Spread? (The Uniformity Trap)

A dataset can look evenly spread across its range while still failing to be a random sample.

Measuring the Spread of Invoice Numbers

- **The Question:** Are invoice numbers clustered in tight clumps or spread smoothly across the 45,223 slots?
- **Cumulative Comparison (ECDF):** We map observed numbers onto $[0, 1]$ and measure the maximum difference D from a straight line.
- **Small Observed Difference:** $D \approx \mathbf{0.012}$ ($n = 22,061$).
- **Intuitive Meaning:** The numbers are spread with high aggregate smoothness across the year—there are no massive multi-thousand-block gaps.

The Critical Pitfall: Even Spread $\neq$ Random Sample

What the Even Spread Tells Us	What It Does NOT Tell Us
Even distribution across span	Whether observations are random draws
Absence of massive block gaps	Whether textbook sampling holds
Descriptive numerical smoothness	Whether observations are exchangeable

Why Standard KS Tests Do Not Apply Here:

- Standard KS tables require fixed, pre-specified endpoints.
- Here, endpoints are estimated from the data ($s_{\min}, s_{\max}$), pinning boundaries at 0 and 1. A formal test would require simulation-based calibration.

Methodological Rule: *Never mistake an evenly spread set of numbers for a random sample. Numerical dispersion across an address span is not random sampling.*

The Core Dilemma: What Exactly Are We Trying to Count?

When identifiers have gaps, there are two completely different population sizes.

Two Conflicting Targets

- **Target A: Address Capacity ($N_{\text{span}} = 45,223$)**
  The total number of integer slots between the first and last invoice ($s_{\max} - s_{\min} + 1$).
- **Target B: Realized Sales ($N_{\text{count}} = 22,061$)**
  The actual count of completed transactions.

Analogy: Estimating the total number of hotel rooms vs. estimating the number of occupied rooms!

What Happens if an Analyst Is Careless?

- **Raw Formula:** Applying $\widehat{N} = m(1 + 1/k) - 1$ directly to raw numbers predicts $\approx \mathbf{593,000}$ (because it assumes numbering starts at 1).
- **Shifted Formula:** Subtracting $s_{\min}$ estimates address capacity $N_{\text{span}} \approx 45,223$, overshooting active sales (22,061) by $+\mathbf{105}\%$ error!
- **The "Oracle" Trap:** Relabeling observed invoices $\{1, 2, \ldots, 22061\}$ recovers active count, but requires knowing all valid numbers in advance!

Fundamental Lesson: You cannot calculate an estimate until you define what you are counting: the total sequence capacity, or the completed items inside it.

Estimand Rule: *Confusing sequence capacity with realized volume produces a 105% estimation disaster.*

The Occupancy Model: Why Incomplete Samples Cannot Count Both

Connecting address capacity to active sales through an intuitive coin-flip model.

The Coin-Flip Occupancy Model

Imagine each numbered slot $j \in \{1, \ldots, N_{\text{span}}\}$ is filled with probability ρ (retention rate):

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho)$$

- N_{span}: Total numbered slots (address capacity).
- N_{count}: Number of slots with completed sales.
- ρ: Proportion of slots that become active sales.

Retrospective Rate on Full Dataset:

$$\widehat{\rho}_{\text{retrospective}} = \frac{22,061}{45,223} \approx 0.4878 \quad (48.8\%)$$

The "Two Knobs" Dilemma in Field Samples

What if we only observe an incomplete field sample ($k \ll N$, e.g., $k = 50$)?

- **The Core Dilemma:** Seeing 20 invoices across a 40-slot sample span could mean:
  1. A huge capacity ($N_{\text{span}} = 100$) with a low completion rate ($\rho = 0.20$), OR
  2. A smaller capacity ($N_{\text{span}} = 40$) with a high completion rate ($\rho = 0.50$).
- Both scenarios produce similar observed sample spacings!
- **Conclusion:** A single incomplete sample cannot disentangle capacity N_{span} from rate ρ without extra information (like audit logs).

Intuitive Takeaway: *The occupancy model explains the math conceptually, but estimating both capacity and void rate from an incomplete sample requires external data.*

Section 3 Verdict: When Does the German Tank Model Work?

Answering the central question: bringing the empirical findings together.

The Scorecard: Comparing Two Worlds

WWII Tanks (Success)	Enterprise Streams (Failure)
Fixed monthly batches	Ongoing streaming transactions
Continuous sequential stamping	51.2% missing integer voids
Independent field captures	Changing seasonal arrival intensity
Target is clear (physical tanks)	Ambiguity: capacity vs. sales count
Statistical error: -4.4%	**Naive formula error:** $+105\%$

Three Golden Rules for Practitioners

- **1. Check How Numbers Are Made:** Determine whether serials come from a static batch or an evolving time-dependent stream.
- **2. Audit the Gaps:** Never assume missing identifiers represent active products without verifying cancellations or database block allocations.
- **3. Define What You Are Counting:** Explicitly choose between sequence capacity (N_{span}) and realized count (N_{count}) before running any estimator.

Final Verdict: The German Tank Problem works brilliantly when physical manufacturing matches textbook assumptions. In modern software and transactional systems, analysts must model the data-generating mechanism first.

Bottom Line: *Statistical elegance cannot rescue a model whose data-generating assumptions are violated.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

The target is the discrete ceiling N, with evidence captured by sample maximum m. Scale-invariant reciprocal prior $\pi(N) \propto 1/n$ yields proper posteriors ($k \geq 1$), respects physical boundaries ($N \geq m$), and matches the large-N pivotal bound.

2. Decision Theory & Rates

Point estimation follows loss geometry: median minimizes absolute error; check loss ($q = 0.9$) builds defensive buffers. Extreme-value order statistics produce accelerated $\mathcal{O}(k^{-1})$ RMSE contraction ($9\times$ reduction for $10\times$ sample).

3. Real-World Reality

Operational data violate textbook assumptions via chronological streaming and integer voids (51.2%). Bernoulli thinning disambiguates address capacity from realized transaction volume.

Central Research Answer: Bayesian inference provides a principled framework for quantifying uncertainty about an unknown discrete population ceiling, selecting estimates under mission-specific decision objectives, and evaluating inferential reliability. Its practical validity in the physical world, however, depends entirely on whether operational data-generation mechanisms satisfy model assumptions.

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